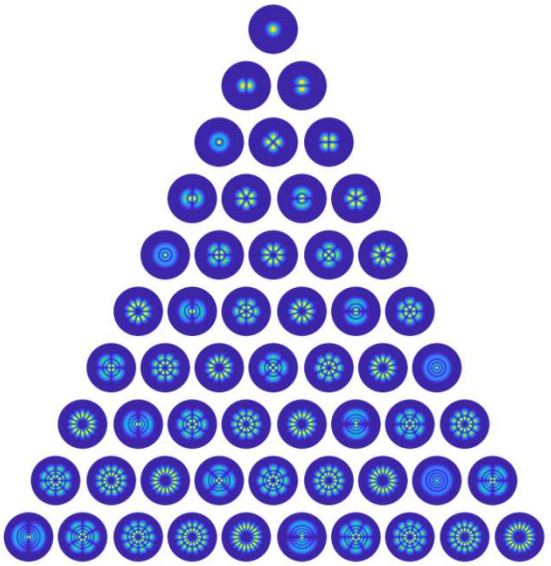
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

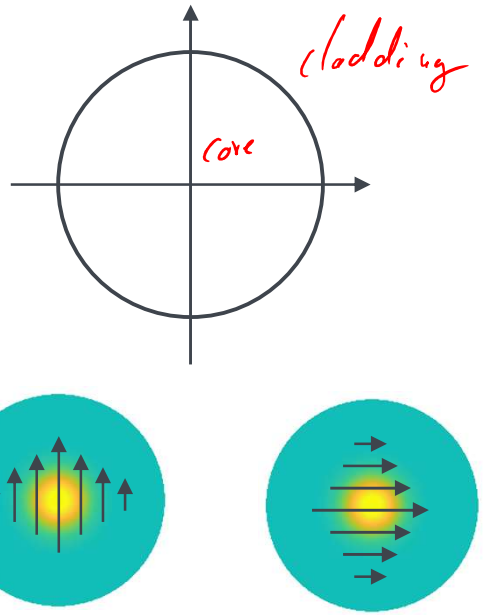
Lecture 13

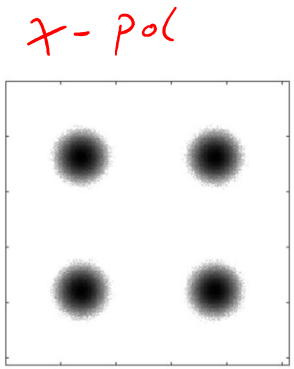


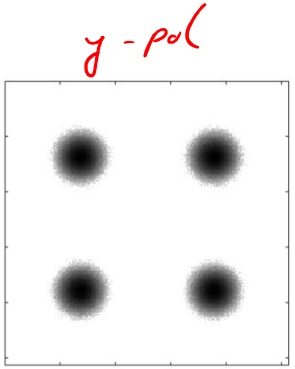
Prof. Dr.-Ing. Georg Rademacher

Polarization multiplex (Pol Mux)

University of Stuttgart
Germany







$$DR: \quad 256\,\mathrm{Bd} \cdot n\,\frac{\mathrm{bit}}{\mathrm{symb}} \cdot \frac{1}{1+OH_{FEC}} \cdot 2\,\mathrm{pol}$$

OSNR in long distance transmission

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Germany

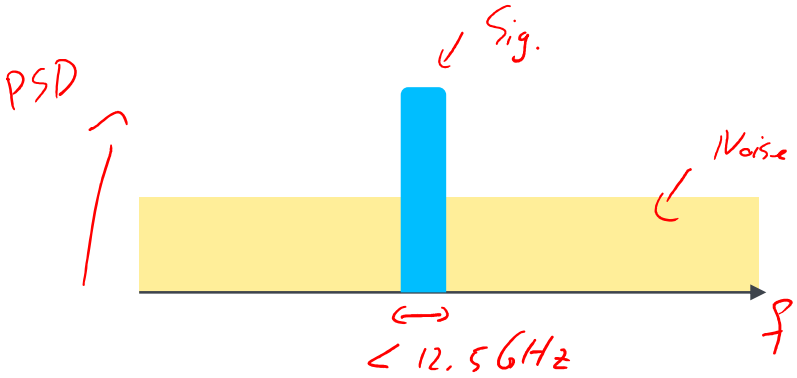
$$OSNR = \frac{P_0}{NF \cdot h \cdot f \cdot \Delta f} = \frac{P_0}{N \cdot G_{ASE} \cdot \Delta f}$$

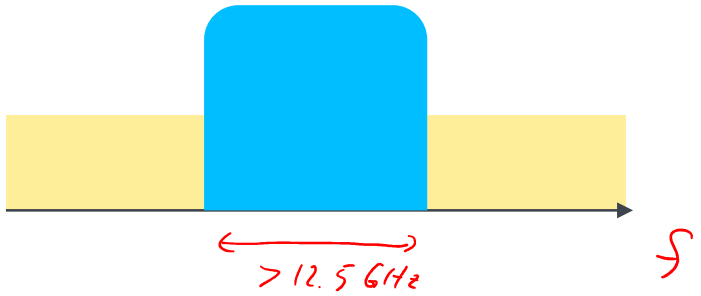
$\Delta f = 12.5\ GHz$ ↓ $\approx 0{,}1\ nm$ bei $1550\ nm$

$$NF_{TOT} \approx N \cdot F_{sog} \cdot Loss_{sog}$$

$$OSNR|_{dB} = 58\,dB + P_0|_{dBm} - NF_{AMP}|_{dB} - \alpha \cdot L_{oa}|_{dB} - 10\log(N)|_{dB}$$

SNR vs. OSNR





- Diff. between wireless and fiber channel

$$SNR = \frac{12.5\,GHz}{R_s} \cdot OSNR$$

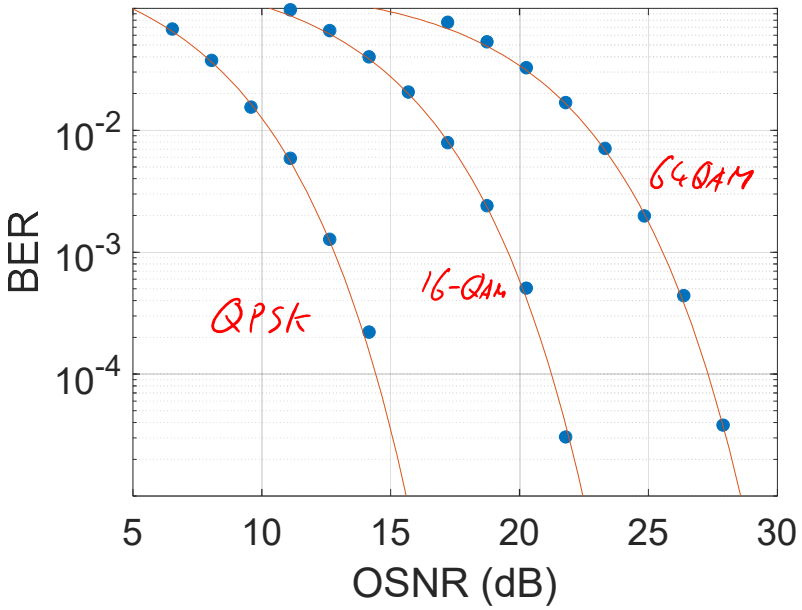
Coherent Signals

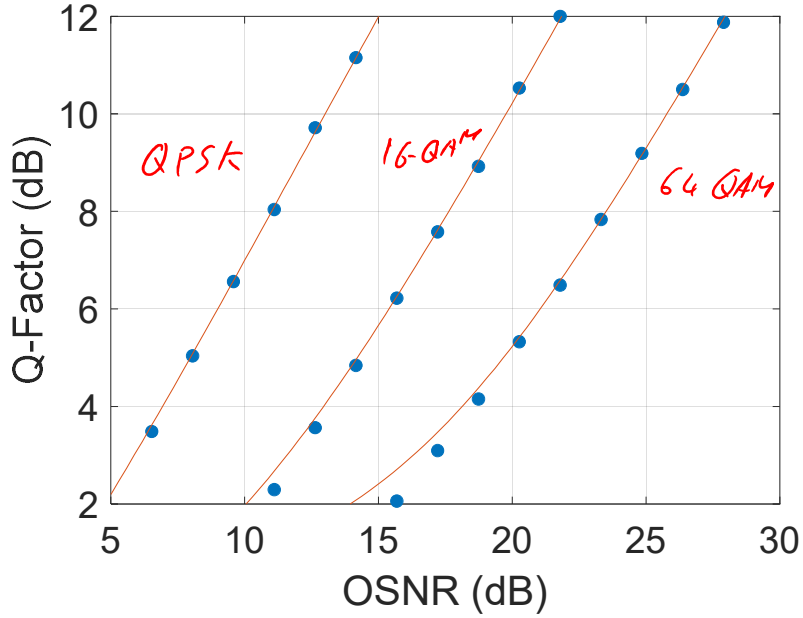
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$$Q_{dB} = 20 \cdot \log_{10}\left(\sqrt{2}\,\mathrm{erfcinv}(2 \cdot BER)\right)$$

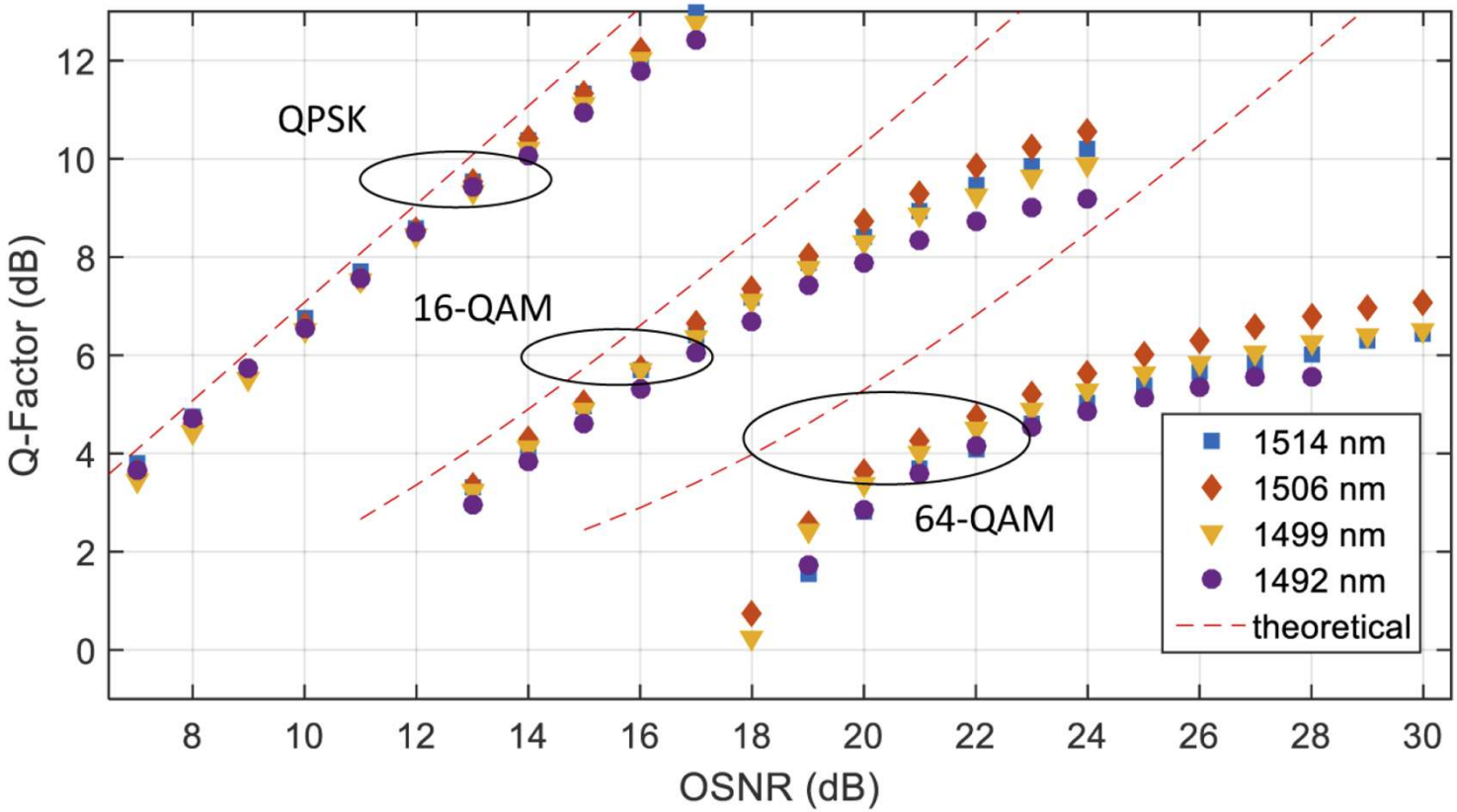
Inverse complementary error function

25 GBd



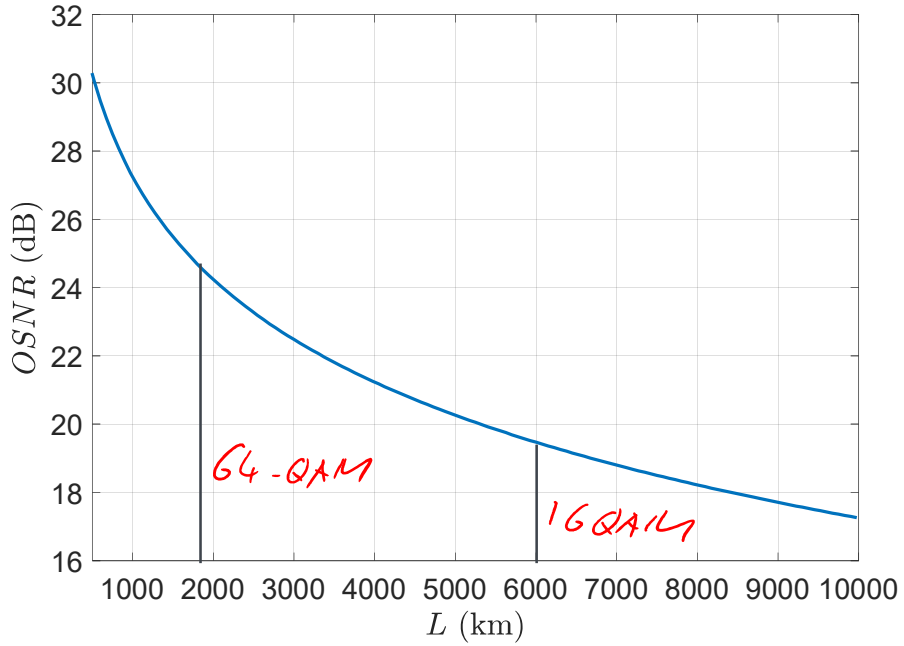


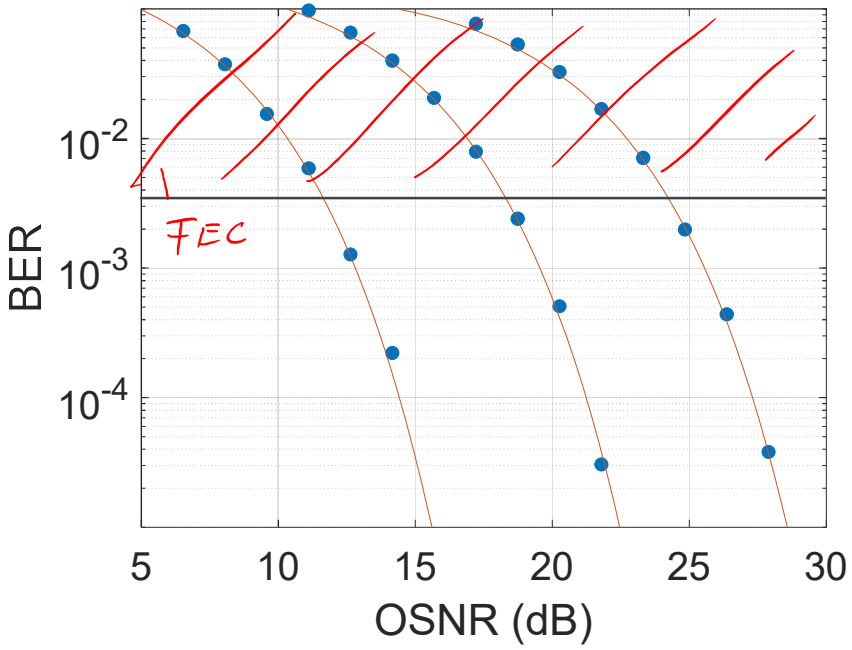
Measured performance



Long distance transmission

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Germany







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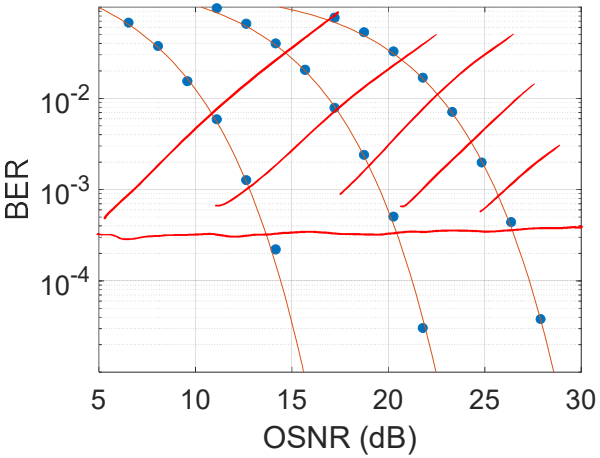
Concept Test

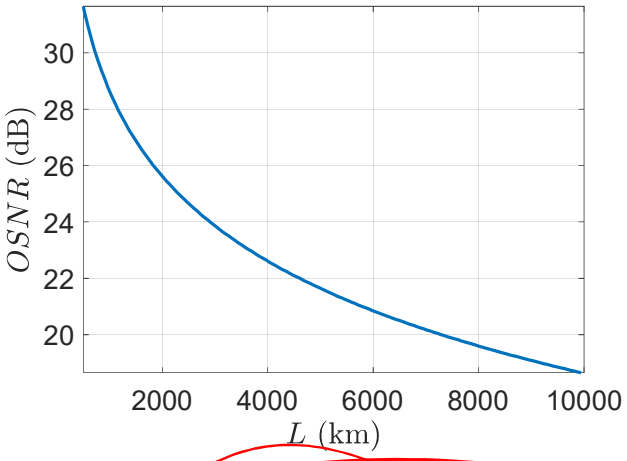
Concept test

DR: $25 \cdot 4\,\frac{bit}{sym} \cdot \frac{1}{(1-10\%)} \cdot 2\,pol = 181.82\,\frac{Gbit}{s}$

University of Stuttgart
Germany

What is the maximum data rate for an FEC limit of $3 \cdot 10^{-4}$ with an overhead of 10% for L = 4500 km at 25 Gbd?





A 45.45 Gbit/s **B** 90.91 Gbit/s **C** 181.82 Gbit/s

https://ilias3.uni-stuttgart.de/vote/DC90

Signal propagation in SMF

University of Stuttgart
Germany

$$\underline{E}(z,t) = A(z,t)\exp(-j\beta_0 z + j\omega_0 t)$$

$$E(z,t) \circ\!\!-\!\!\bullet\, \underline{E}(z,j\omega)$$

$$\underline{E}(z,j\omega) = \underline{E}(z=0,j\omega)\exp(-j\beta(\omega)\cdot z)$$

$\beta(\omega)$ → develop into Taylor-Series around ω_0

$$\beta(\omega) = \beta_0 + \beta_1(\omega-\omega_0) + \frac{1}{2}\beta_2(\omega-\omega_0)^2 + \frac{1}{6}\beta_3(\omega-\omega_0)^3 + \dots$$

Pulse propagation in SMF

University of Stuttgart
Germany

$\beta_0 = \beta(\omega)$: phase constant

$\beta_1 = \frac{d\beta}{d\omega} = \tau$: group delay

$\beta_2 = \frac{d^2\beta}{d\omega^2} = \frac{d\tau}{d\omega} = -\frac{d\tau}{\lambda}\left(\frac{\lambda^2}{2\pi c}\right)$: chromatic dispersion

$\beta_3 = \frac{d^3\beta}{d\omega^3} = \frac{d}{d\omega}\left(\frac{d\tau}{d\omega}\right)$: dispersion slope

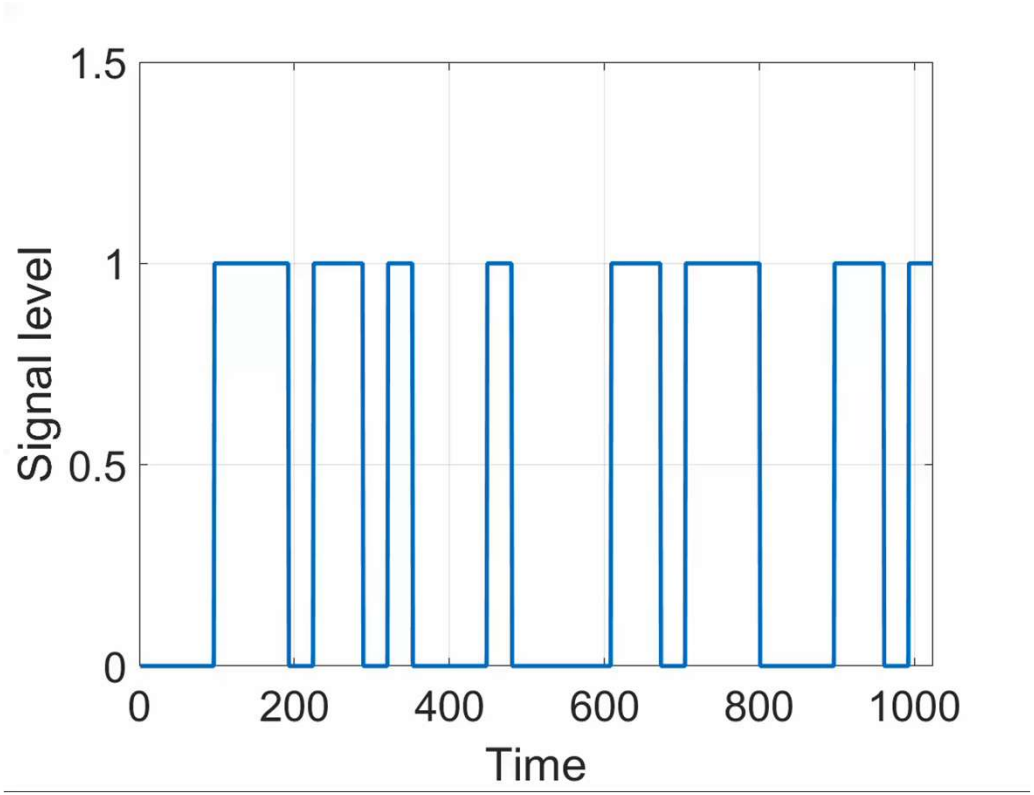
Pulse propagation in SMF

$$\frac{\partial A}{\partial z} + \beta_1 \frac{\partial A}{\partial t} + j\frac{\beta_2}{2}\frac{\partial^2 A}{\partial t^2} - \frac{\beta_3}{6}\frac{\partial^3 A}{\partial t^3} = 0$$

$$t' = t - \beta_1 \cdot z \quad , \quad z' = z$$

$$\frac{\partial A}{\partial z'} + j\frac{\beta_2}{2}\frac{\partial^2 A}{\partial t'^2} - \frac{\beta_3}{6}\frac{\partial^3 A}{\partial t'^3} = 0$$

Pulse propagation in SMF: Chromatic dispersion





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Bonus Test

Bonus test

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Which of the following three effects is not fundamentally limiting the performance of an optical transmission link?

- **A** Attenuation
- **B** Noise
- **C** Dispersion

ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_C7N9

Nonlinear signal distortions

University of Stuttgart
Germany

$$n' = n + \bar{n}_2 \left(\frac{P}{A_{eff}} \right)$$

$A_{eff} \approx 80\,\mu m^2$

$\bar{n}_2 \approx 2.6 \times 10^{-20} \frac{m^2}{W}$

$$\frac{\partial A}{\partial z} + j \frac{\beta_2}{2} \frac{\partial^2 A}{\partial t^2} = j\gamma |A|^2 A - \frac{\alpha}{2} A$$

$$\gamma = \frac{\bar{n}_2 \cdot \omega}{c \cdot A_{eff}} \quad , \quad \gamma = \frac{1.31}{W \cdot km} \frac{80\,\mu m^2}{A_{eff}}$$

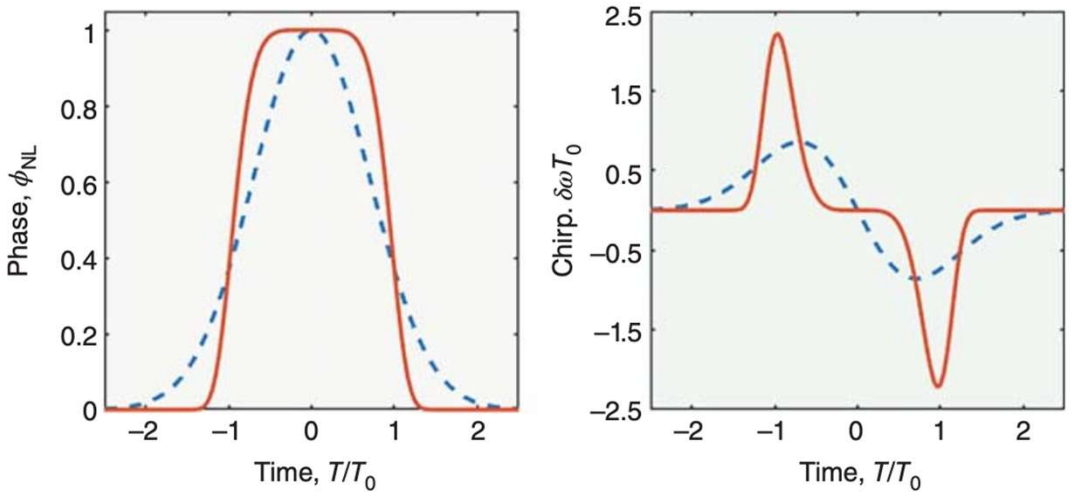
Self-phase modulation

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Germany

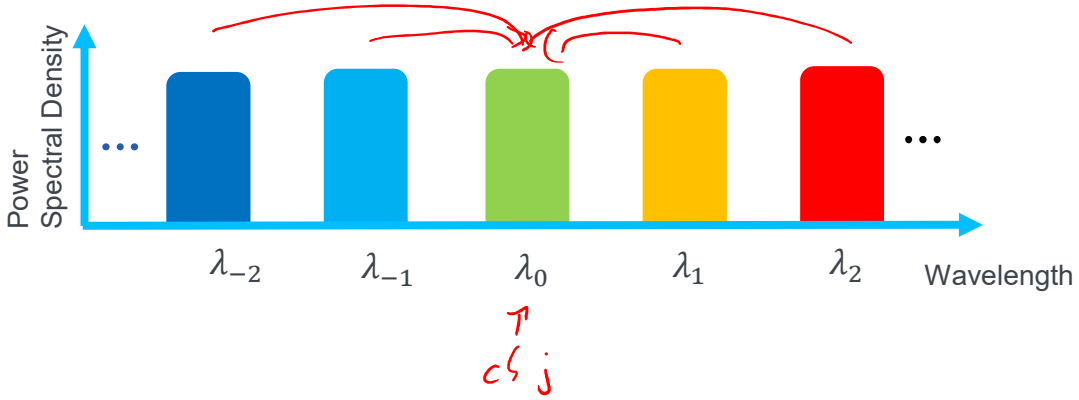
$$\beta' = \beta + k_0 \cdot n_2 \frac{P}{A_{eff}} = \beta + \gamma \cdot P$$

$$\phi_{NL} = \int_0^L (\beta' - \beta)\, dz = \int_0^L P(z)\, dz = \gamma \cdot P_{in} \cdot L_{eff}$$

$$L_{eff} = [1 - \exp(-\alpha L)] / \alpha$$

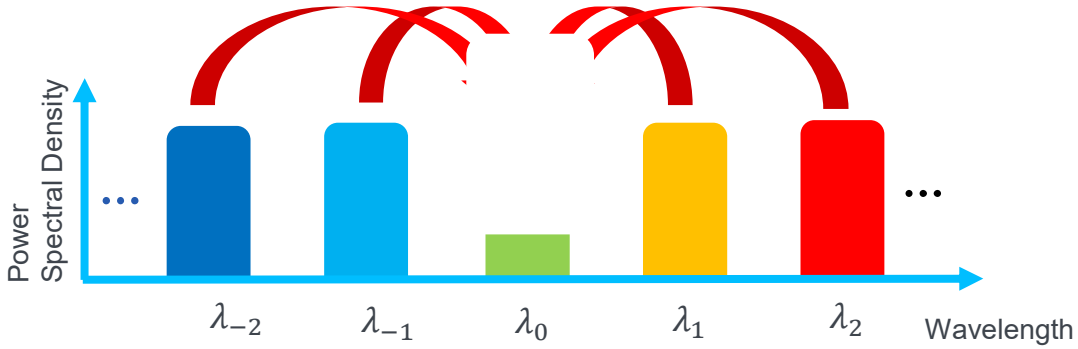


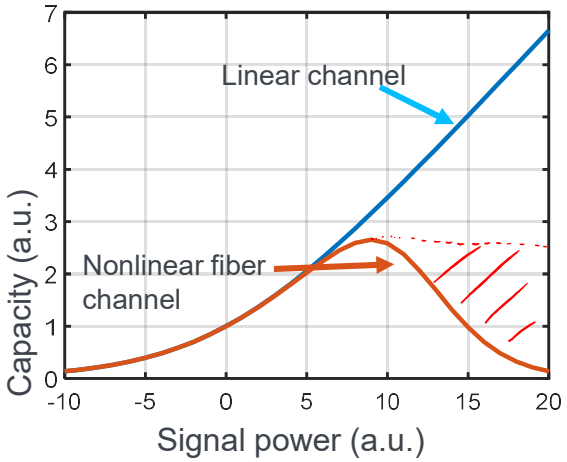
Cross-phase modulation



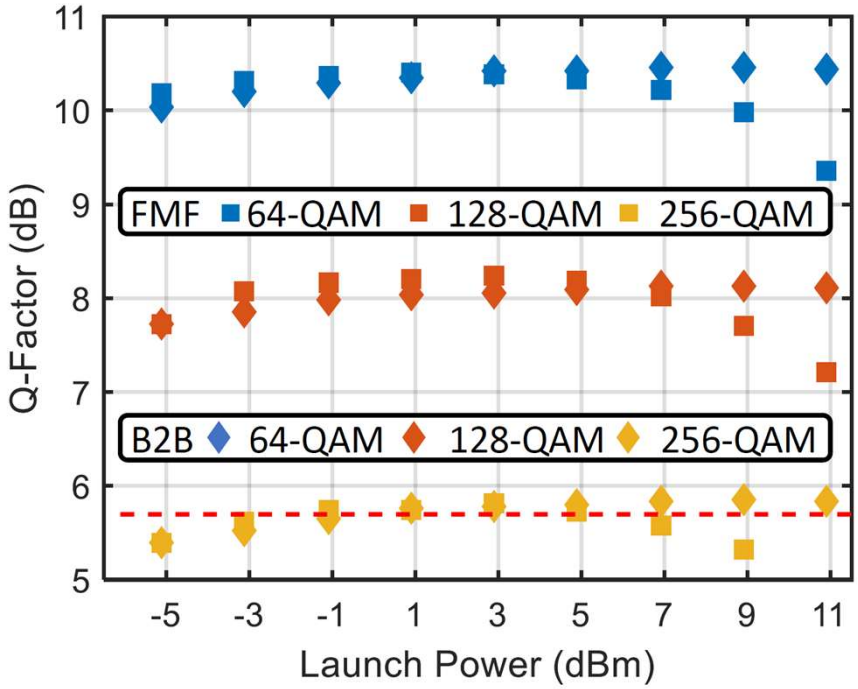
$$\phi_{NL} = \gamma \cdot L_{eff} \left(P_j + 2 \sum_{m \neq j} P_m \right)$$

Nonlinear signal distortions

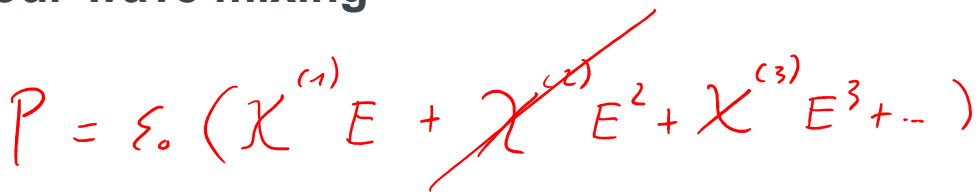




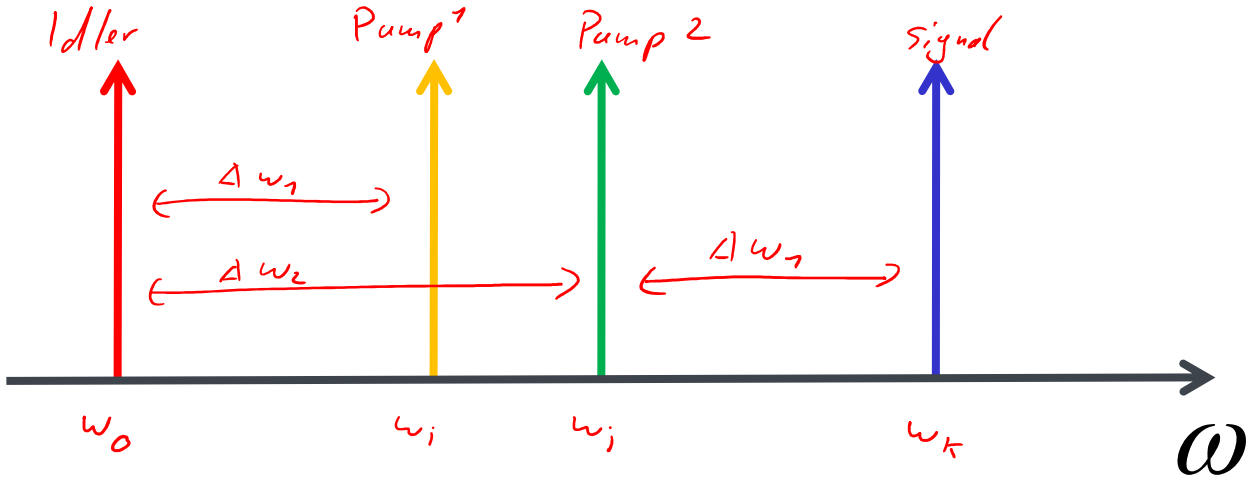
Nonlinear signal distortions

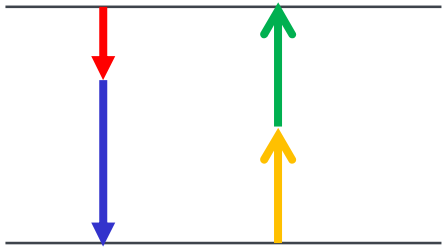


Four-wave mixing



Conservation of energy





Four-wave mixing

$$\omega_0 = \omega_i + \omega_j - \omega_k$$

$$\Delta\beta = -\beta(\omega_0) + \beta(\omega_i) + \beta(\omega_j) - \beta(\omega_k)$$

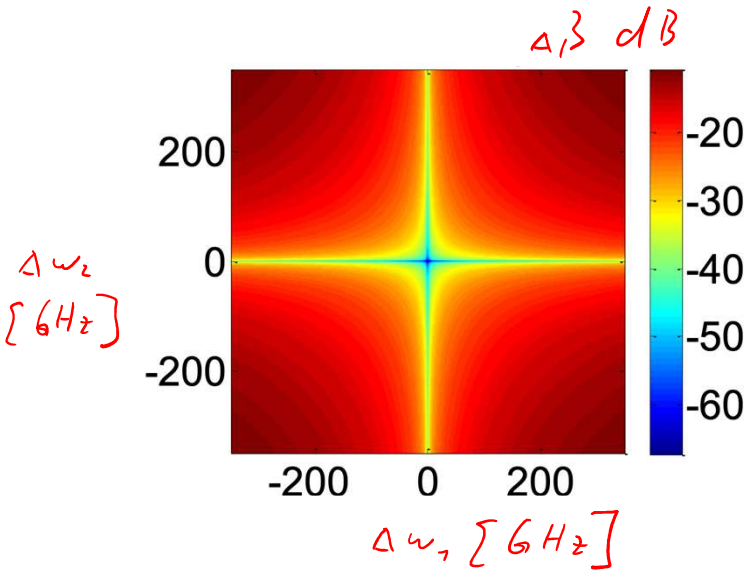
$$\beta(\omega) = \beta_0 + \beta_1(\omega - \omega_0) + \frac{1}{2}\beta_2(\omega - \omega_0)^2 + \dots$$

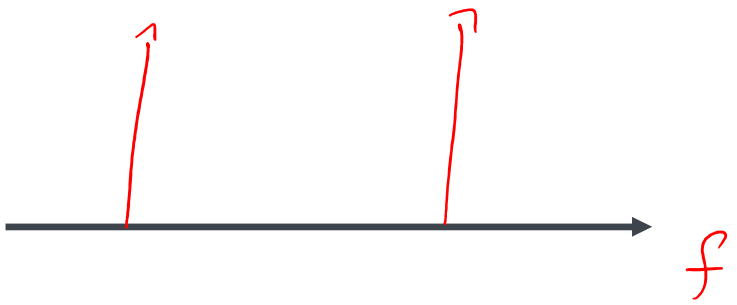
conservation of momentum

$$\Delta\beta(\Delta\omega_1, \Delta\omega_2) = \beta_2(\Delta\omega_1 - \omega_0)(\Delta\omega_2 - \omega_0)$$

Four-wave mixing

University of Stuttgart
Germany





Nonlinear optical signal-to-noise ratio

$$OSNR_{NL} = \frac{P_0}{N\,(G_{ASE} + G_{NLI})\cdot 12.5\,GHz}$$

JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 24, DECEMBER 15, 2012 3857

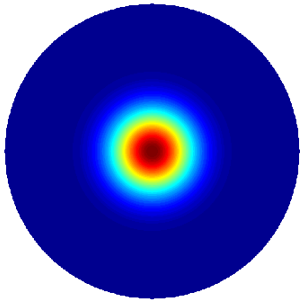
The GN Model of Non-Linear Propagation in Uncompensated Coherent Optical Systems

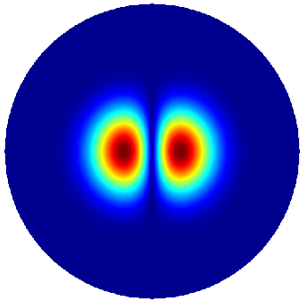
Pierluigi Poggiolini, *Member, IEEE*

(Invited Paper)

Four-wave mixing

University of Stuttgart
Germany

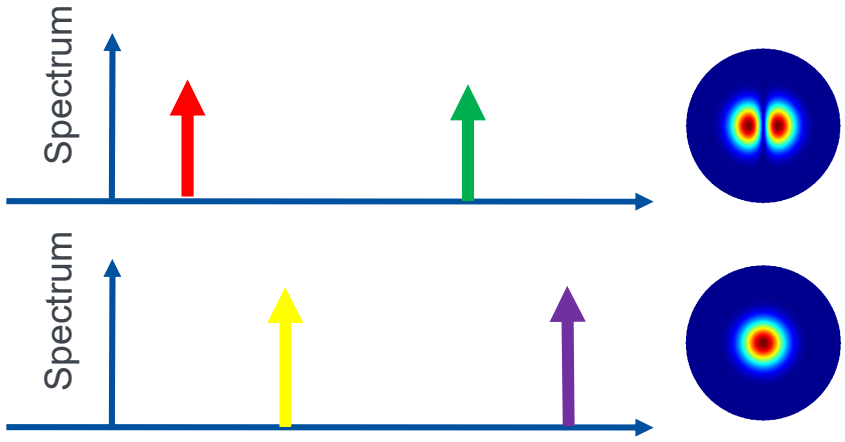


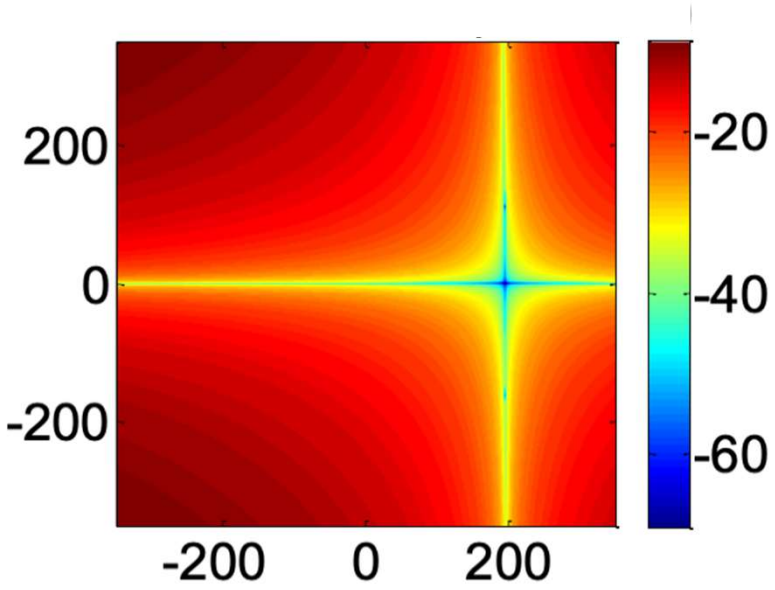


Intermodal four-wave mixing

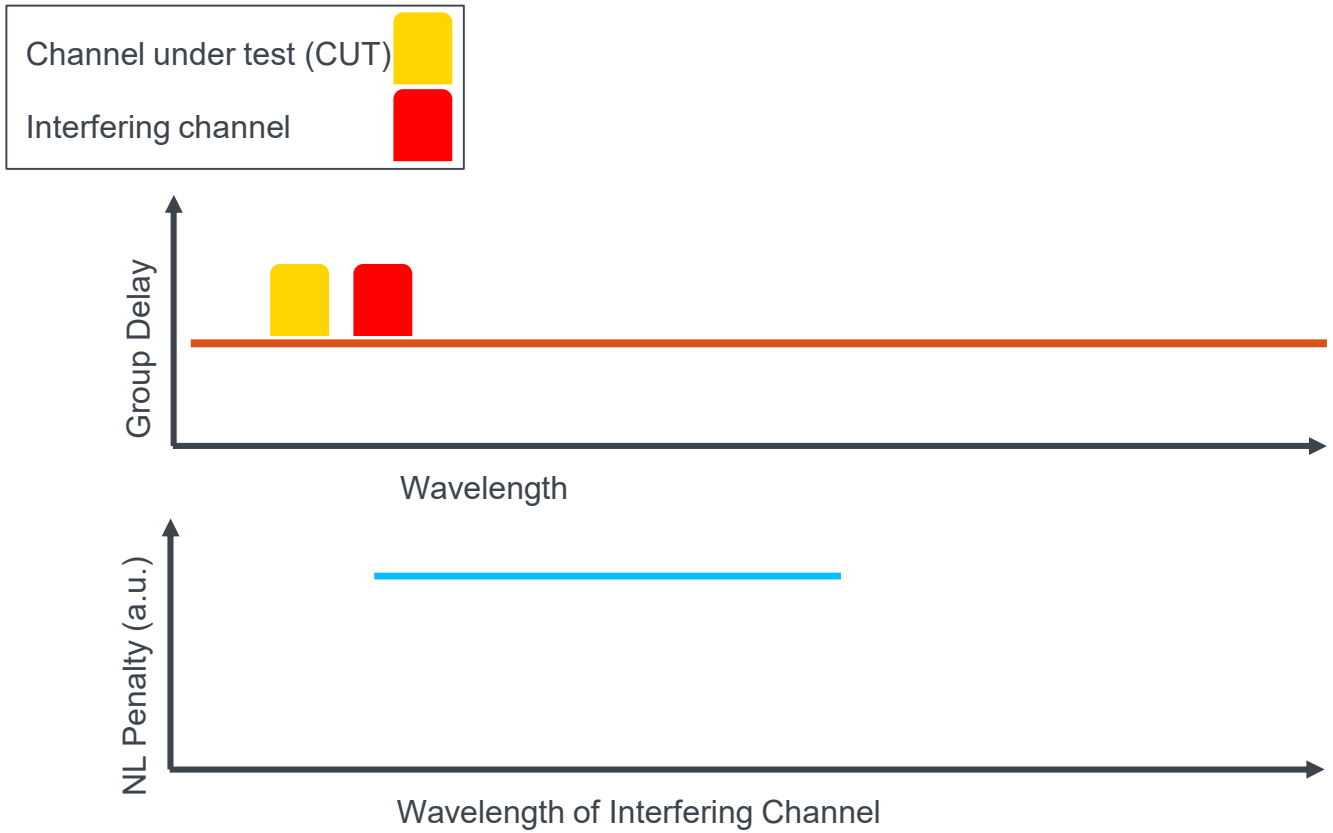
Intermodal four-wave mixing

University of Stuttgart
Germany

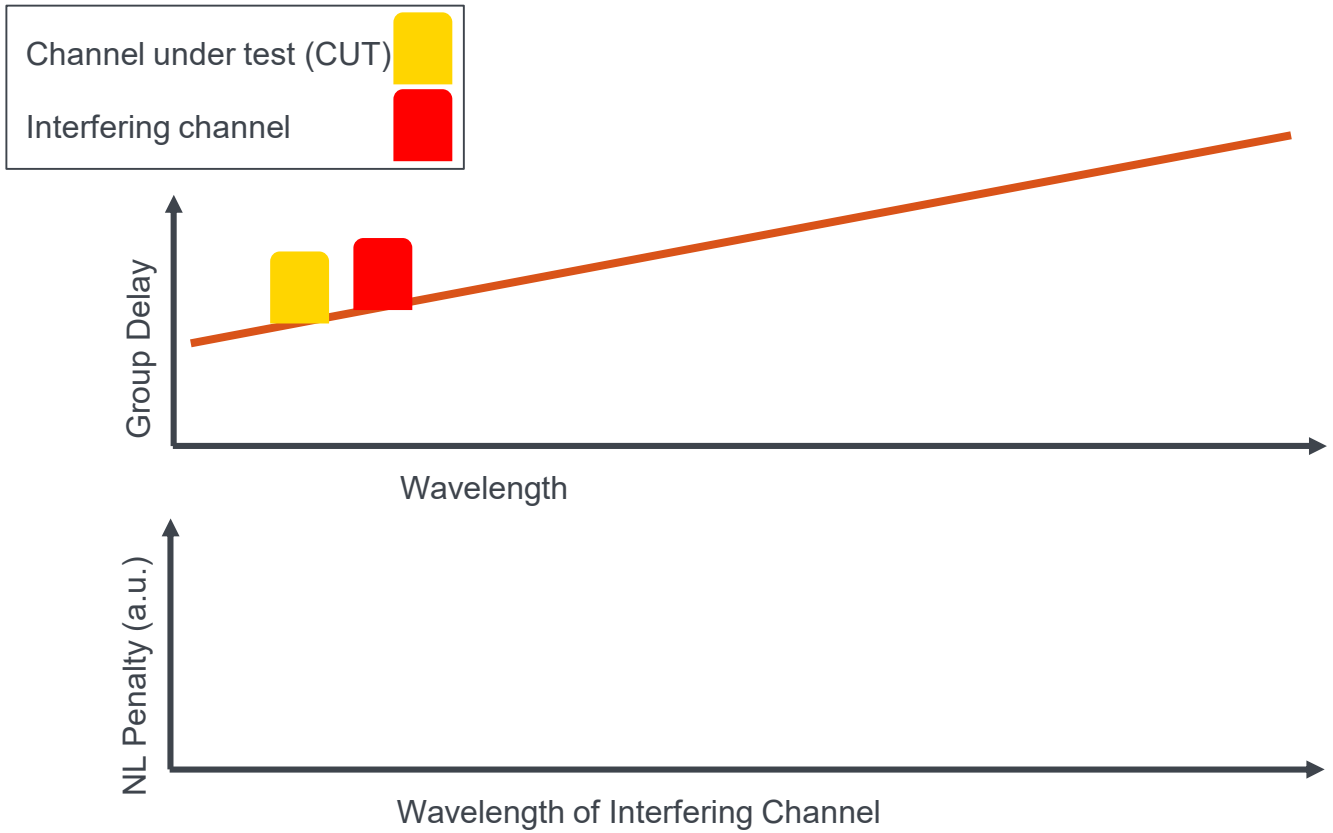




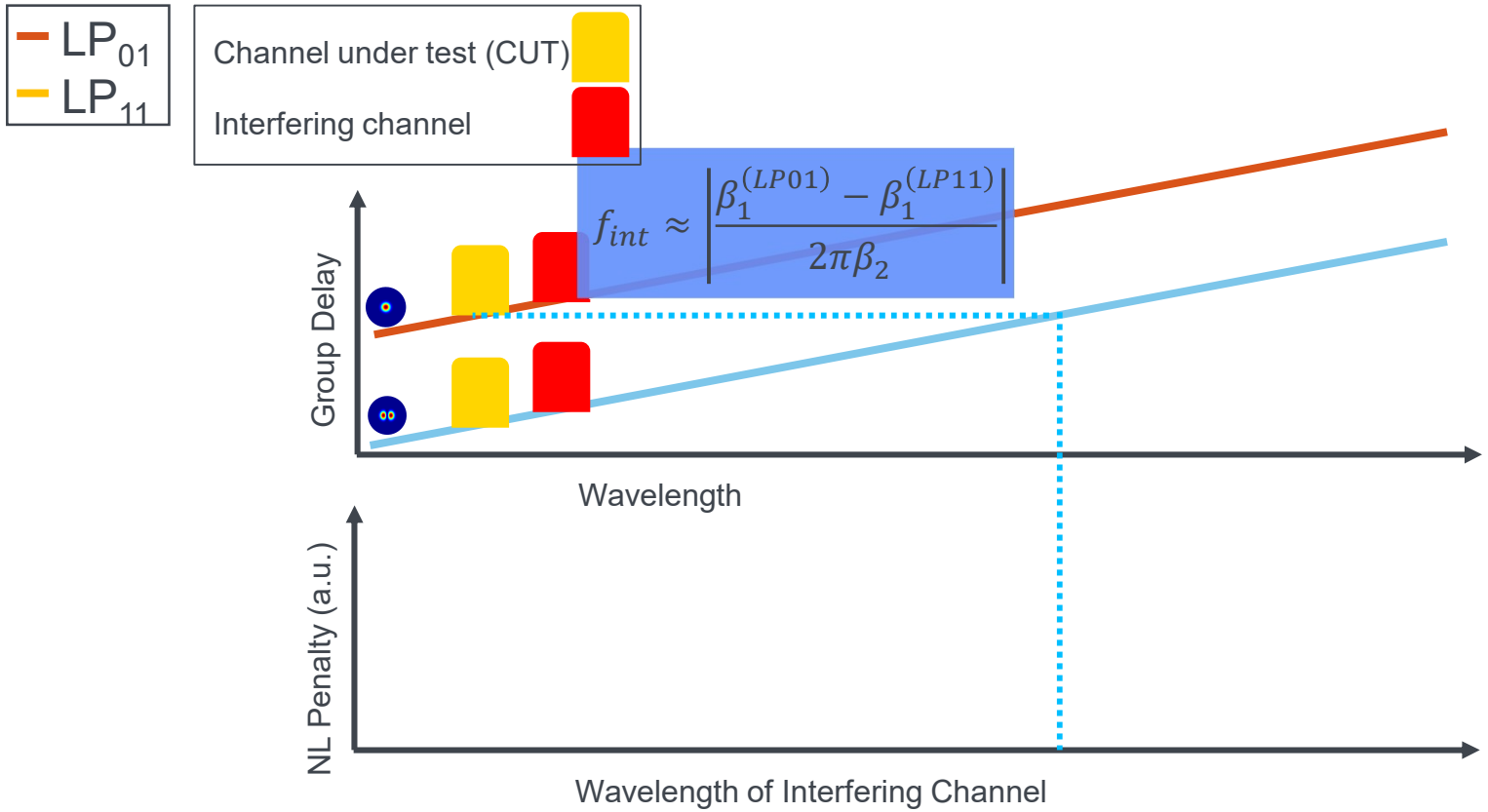
Nichtlineare Signalverzerrung in einer nicht dispersiven Single-Mode Faser



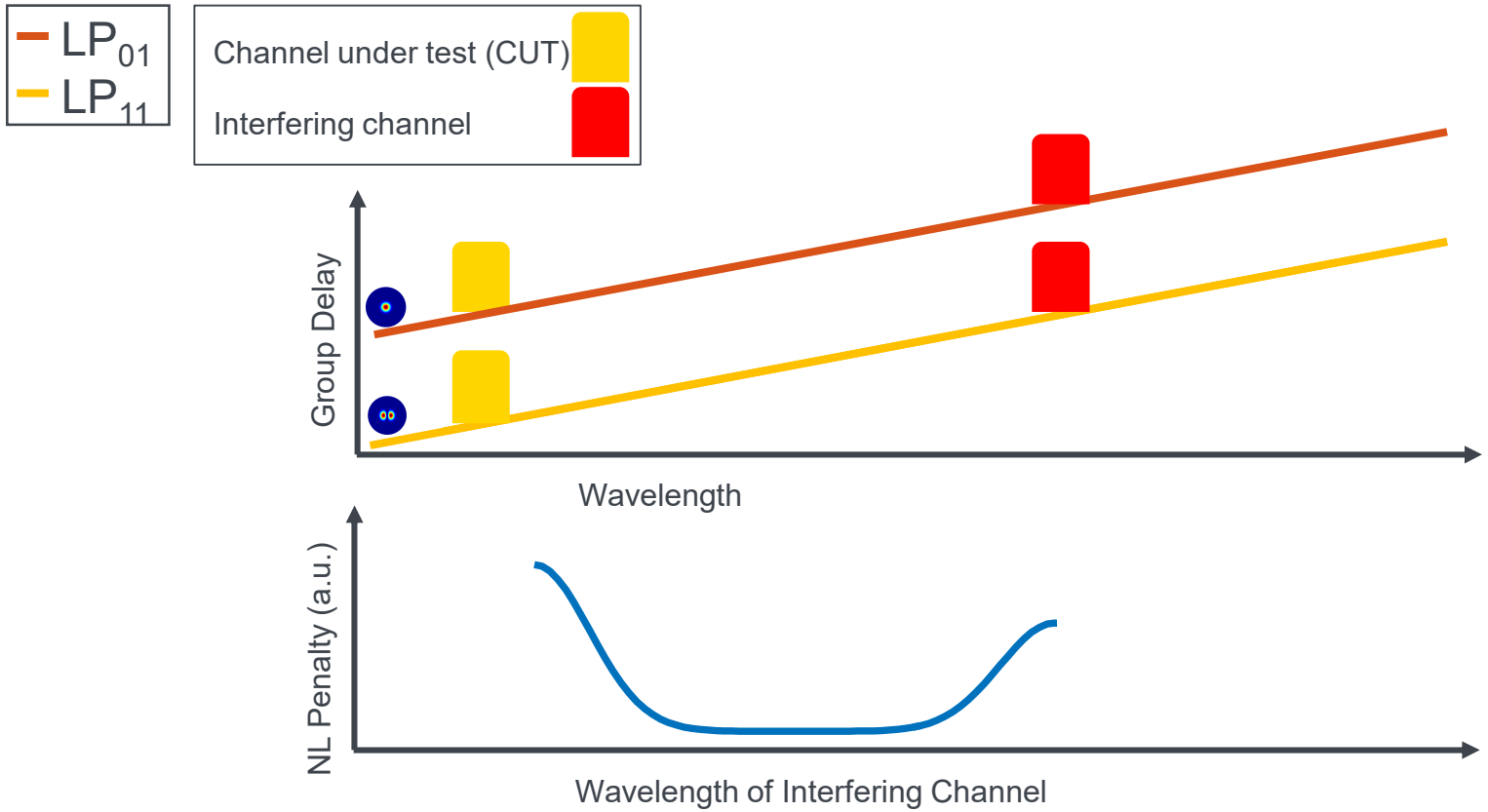
Nichtlineare Signalverzerrung in einer standard Single-Mode Faser



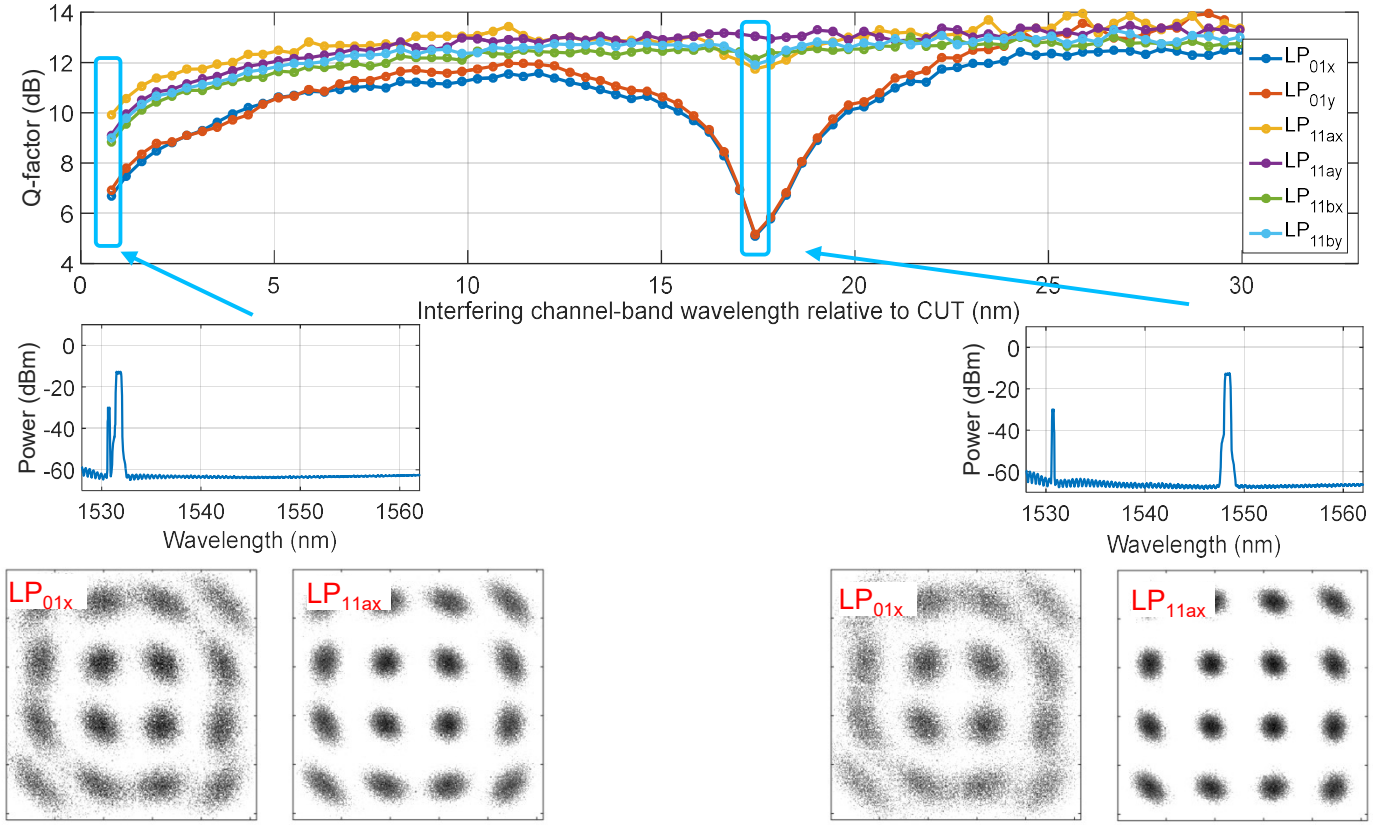
Nichtlineare Signalverzerrung in einer Few-Mode Faser



Nichtlineare Signalverzerrung in einer Few-Mode Faser



Experimentale Demonstration



G. Rademacher et al., “Investigation of Intermodal Nonlinear Signal Distortions in Few-Mode Fiber Transmission”, Journal of Lightwave Technology vol. 37, 2019

University of Stuttgart
Institute of electrical and optical communications

Thank you!



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